

FSPP0034 PHYSICS II

Tutorial 4

Chapter 6 & 7

$$\mathcal{E} = -N \frac{d\Phi}{dt} = -2 \left(\frac{38 - (-58)}{0.42} \right) = -467 \text{ V}$$

1. The magnetic flux through a coil of wire containing two loops changes at a constant rate from -58 Wb to +38 Wb in 0.42 s. What is the emf induced in the coil?

[Ans: 460 V]

2. A 22.0-cm-diameter loop of wire is initially oriented perpendicular to a 1.5-T magnetic field. The loop is rotated so that its plane is parallel to the field direction in 0.20 s. What is the average induced emf in the loop?

$$\mathcal{E} = -\frac{d\Phi}{dt} = \frac{BA \Delta \cos \theta}{t} = \frac{1.5 (0.038) \cos 0}{0.2} = -0.285 \text{ V}$$

[Ans: 0.29 V]

3. A 16-cm-diameter circular loop of wire is placed in a 0.50-T magnetic field.

- (a) When the plane of the loop is perpendicular to the field lines, what is the magnetic flux through the loop?

$$\Phi = BA = 0.50 (\pi \times 0.08^2) = 0.01 \text{ Tm}^2$$

- (b) The plane of the loop is rotated until it makes a 35° angle with the field lines. What is the angle, θ for this situation?

$$90^\circ - 35^\circ = 55^\circ$$

- (c) What is the magnetic flux through the loop at this angle?

$$\Phi = 0.01 (\cos 55^\circ) = 5.74 \times 10^{-3} \text{ Tm}^2$$

[Ans: (a) 1.0×10^{-2} Wb

(b) 55°

(c) 5.8×10^{-3} Wb

4. A circular loop in the plane of the paper lies in a 0.75-T magnetic field pointing into the paper. If the loop's diameter changes from 20.0 cm to 6.0 cm in 0.50 s,

- (a) what is the direction of the induced current, *into the page*

$$\mathcal{E} = -\frac{B \Delta A}{t}$$

- (b) what is the magnitude of the average induced emf,

$$= \frac{0.75 (\cos 90^\circ) (2.83 \times 10^{-3} - 3.14 \times 10^{-3})}{0.5}$$

- (c) if the coil resistance is 2.5 Ω , what is the average induced current?

$$\mathcal{E} = IR$$

$$0.043 = I(2.5)$$

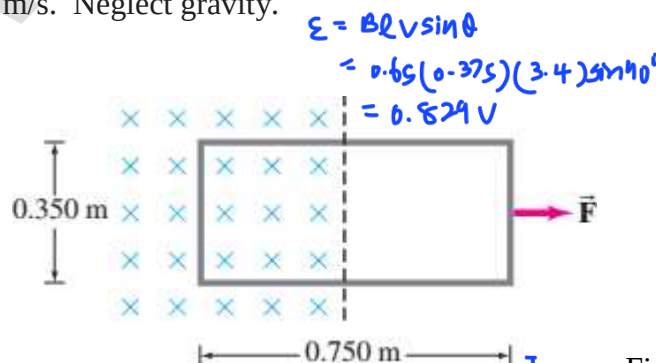
$$I = 0.0172 \text{ A}$$

[Ans: (a) 4.3×10^{-2} V

(b) 1.7×10^{-2} A]

$$= -0.043 \text{ V}$$

5. Part of a single rectangular loop of wire with dimensions shown in Fig. 1 is situated inside a region of uniform magnetic field of 0.650 T. The total resistance of the loop is 0.280 Ω . Calculate the force required to pull the loop from the field (to the right) at a constant velocity of 3.40 m/s. Neglect gravity.



[Ans: 0.628 N]

$$I = \frac{0.829}{0.280} = 2.96 \text{ A}$$

$$F = 2.96 (0.375) (0.65) \sin 90 = 0.634 \text{ N}$$

$$F = ILB, \mathcal{E} = BLv, \mathcal{E} = \frac{I}{R}$$

$$= \frac{\mathcal{E}}{R} LB$$

$$= \frac{(BLv)(BL)}{R}$$

$$= \frac{B^2 L^2 v}{R}$$

Figure 1

$$\mathcal{E} = -\frac{d\Phi}{dt} \Rightarrow \mathcal{E} = -\frac{(0-0.4)(0.049)\cos 0^\circ}{120 \times 10^{-3}} = 0.164 \text{ V} \quad \begin{matrix} I = 1.1 \times 10^{-3} \text{ A} \\ \mathcal{E} = IR \end{matrix}$$

6. A 25-cm-diameter circular loop of wire has a resistance of $150 \, \Omega$. It is initially in a 0.40-T magnetic field, with its plane perpendicular to $\vec{B}$ but is removed from the field in 120 ms. Calculate the electric energy dissipated in this process. $E = Pt$

$$= I^2 R t = 2.178 \times 10^{-5} \text{ J} \quad [\text{Ans: } 2.1 \times 10^{-5} \text{ J}]$$

7. In Fig. 2, the rod moves to the right with a speed of 1.3 m/s and has a resistance of $2.5 \, \Omega$.

The rail separation is $l = 25.0 \text{ cm}$. The magnetic field is 0.35 T, and the resistance of the U-shaped conductor is $25.0 \, \Omega$ at a given instant. Calculate:

(a) the induced emf, $\mathcal{E} = Blv \sin \theta = 0.35(0.25)(1.3) \sin 90^\circ = 0.114 \text{ V}$

(b) the current in the conductor, $\mathcal{E} = IR$, $I = \frac{0.114}{27.5} = 4.13 \times 10^{-3} \text{ A}$

- (c) the external force needed to keep the rod's velocity constant at that instant.

$$F = IlB \sin \theta = 4.1 \times 10^{-3}(0.25)(0.35) = 3.6 \times 10^{-4} \text{ N} \quad [\text{Ans: (a) } 0.11 \text{ V} \quad (\text{b}) 4.1 \text{ mA} \quad (\text{c}) 0.36 \text{ mN}]$$

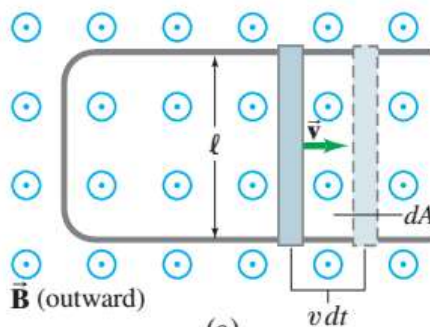


Figure 2

8. A simple generator is used to generate a peak output voltage of 24.0 V. The square armature consists of windings that are 5.15 cm on a side and rotates in a field of 0.420 T at a rate of 60.0 rev/s. How many loops of wire should be wound on the square armature?

$$\omega = \frac{60 \text{ rev}}{1 \text{ s}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = 120\pi \quad 24 = N(0.42)(2.65 \times 10^{-3})(120\pi) \quad [\text{Ans: } 57.2 \text{ loops}]$$

$$N = 57.19 \text{ loops}$$

9. A simple generator has a 480-loop square coil 22.0 cm on a side. How fast must it turn in a 0.550-T field to produce a 120-V peak output? $\mathcal{E}_{\text{max}} = NBA\omega$

$$120 = 480(0.55)(0.0484)\omega \quad [\text{Ans: } 1.49 \text{ rev/s}]$$

$$\omega = 9.39 \text{ rad/s} = 1.49 \text{ rev/s}$$

10. Neon signs require 12 kV for their operation. To operate from a 240-V line, what must be the ratio of secondary to primary turns of the transformer? What would the voltage output be if the transformer were connected backward?

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{12000}{240} = 50, \quad \frac{V_s}{240} = \frac{1}{50}, \quad V_s = 4.8 \text{ V} \quad [\text{Ans: (a) } 50 \quad (\text{b}) 4.8 \text{ V}]$$

11. The output voltage of a 75-W transformer is 12 V, and the input current is 22 A.

(a) Is this a step-up or a step-down transformer?

(b) By what factor is the voltage multiplied?

$$\text{(a) } P = IV \quad 75 = 22V \quad \text{step-up} \quad \text{(b) } \frac{V_s}{V_p} = \frac{12}{3.41} = 3.52 \quad [\text{Ans: (a) step down} \quad (\text{b}) 3.5]$$

$$V_p = 3.41 \text{ V}$$

12. A 2.44 m long coil containing 225 loops is wound on an iron core (average $\mu = 1850 \mu_0$) along with a second coil of 115 loops. The loops of each coil have a radius of 2.00 cm. If

$$M = \frac{\mu_0 N_1 N_2 A}{l} = \frac{4\pi \times 10^{-7} (225)(115)(\pi \times 0.02^2)}{2.44} \times 1850 = 0.031 \text{ H} \quad \mathcal{E} = -M \frac{dI}{dt} = -0.031 \left(\frac{0-12}{98 \times 10^{-3}} \right) = 3.8 \text{ V}$$

the current in the first coil drops uniformly from 12.0 A to zero in 98.0 ms, determine: = 3.8 V

- (a) the mutual inductance M ;
(b) the emf induced in the second coil.

[Ans: (a) $3 \times 10^{-2} \text{ H}$ (b) 3.79 V]

13. If the current in a 280-mH coil changes steadily from 25.0 A to 10.0 A in 360 ms, what is the magnitude of the induced emf?

$$\mathcal{E} = -280 \times 10^{-3} \left(\frac{10-25}{360 \times 10^{-3}} \right) = 11.67 \text{ V} \quad [\text{Ans: } 12 \text{ V}]$$

14. What is the inductance of a coil if the coil produces an emf of 2.50 V when the current in it changes from -28.0 mA to +25 mA in 12.0 ms?

$$2.50 = -M \left(\frac{25 - (-28) \times 10^{-3}}{12 \times 10^{-3}} \right), M = 0.566 \text{ H} \quad [\text{Ans: } 0.57 \text{ H}]$$

15. An air-filled cylindrical inductor has 2800 turns, and it is 2.5 cm in diameter and 21.7 cm long.

$$L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} (2800^2) (4.91 \times 10^{-4})}{0.217} = 0.022 \text{ H}$$

- (a) What is its inductance?

- (b) How many turns would you need to generate the same inductance if the core were filled with iron of magnetic permeability 1200 times that of free space?

$$0.022 = \frac{1200(4\pi \times 10^{-7})(N^2)(4.91 \times 10^{-4})}{0.217}, N^2 = 6447, N = 80.3 \text{ turns} \quad [\text{Ans: (a) } 0.022 \text{ H (b) } 81 \text{ turns}]$$

16. It takes 2.56 ms for the current in an LR circuit to increase from zero to 0.75 its maximum value. Determine: (a) the time constant of the circuit, $I = I_0(1 - e^{-\frac{t}{\tau}})$

- (b) the resistance of the circuit if $L = 31.0 \text{ mH}$. $I = 0.75 I_0, 0.75 = 1 - e^{-\frac{2.56 \times 10^{-3}}{\tau}}$

$$1.85 \times 10^{-3} = \frac{31 \times 10^{-3}}{R}, R = 16.8 \Omega \quad [\text{Ans: (a) } 1.85 \text{ ms (b) } 17 \Omega]$$

17. If the magnetic field in a traveling EM wave has a peak magnitude of 12.5 nT, what is the peak magnitude of the electric field? $\tau = 1.847 \times 10^{-3} \text{ s}$

[Ans: 3.75 V/m]

18. What is the frequency of a microwave whose wavelength is 1.50 cm?

[Ans: $2.00 \times 10^{10} \text{ Hz}$]

19. An electromagnetic wave has an electric field given by

$$\vec{E} = \hat{i}(225 \text{ V/m}) \sin[(0.077 \text{ m}^{-1})z - (2.3 \times 10^7 \text{ rad/s})t].$$

- (c) What are the wavelength and frequency of the wave?
(d) Write down an expression for the magnetic field.

$$[\text{Ans: (a) } 82 \text{ m, } 3.7 \text{ MHz (b) } \vec{B} = \hat{j}(7.50 \times 10^{-7} \text{ T}) \sin[(0.077 \text{ m}^{-1})z - (2.3 \times 10^7 \text{ rad/s})t]]$$

$$17) \quad B = 12.5 \times 10^{-9}$$

$$v = \frac{E}{B}$$

$$3 \times 10^8 = \frac{E}{12.5 \times 10^{-9}}$$

$$E = 3.75 \text{ V/m}$$

$$18) \quad v = f\lambda$$

$$3 \times 10^8 = f(0.015)$$

$$f = 2 \times 10^{10} \text{ Hz}$$

$$19) \quad k = 0.077$$

$$\omega = 2.3 \times 10^7$$

$$E = 225$$

$$k = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{2\pi}{k}$$

$$= 81.61 \text{ m}$$

$$v = f\lambda$$

$$3 \times 10^8 = f(81.61)$$

$$f = 3.67 \times 10^6 \text{ Hz}$$

$$v = \frac{E}{B}$$

$$B = \frac{225}{3 \times 10^8}$$

$$\vec{B} = j \sin(kx - \omega t)$$

$$= 7.5 \times 10^{-7} j \sin(0.077z - 2.3 \times 10^7 t)$$